

DAWOOD
UNIVERSITY
OF ENGINEERING & TECHNOLOGY

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI
MIDSEMESTER EXAMINATION OF 1ST SEMESTER, 1ST YEAR
FALL SEMESTER 2025 OF BS ARTIFICIAL INTELLIGENCE
SECTIONS (A1, A2 & A3)

APPLIED PHYSICS: (THEORY)

DATED: 29-10-2025

TIME ALLOWED 90 MINUTES

MAX. MARKS 30

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 04 questions.
2. Use of Calculator is allowed. Use of mobile phone as a calculator is not allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink (Blue or black). For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationery is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned to invigilator before leaving exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS AND EXPLAIN EACH STEP FOLLOWED

Q.NO. 1 (A) Define briefly the following with examples. (CLO1, PLO1, 4.5 MARKS)

- (i) Scientific notation and significant figures.
- (ii) Derived units and supplementary units.
- (iii) Dimensions of physical quantities

(B) Show that following equations are dimensionally correct. (CLO3, PLO3, 3 MARKS)

- (i) $T = 2\pi \sqrt{\frac{l}{g}}$ (g is acceleration due to gravity), (ii) $F = \frac{mv^2}{r}$ (Centripetal force)

Q.NO. 2 (A) Define product of two vector and show that $|\vec{A} \times \vec{B}|^2 + |\vec{A} \cdot \vec{B}|^2 = |\vec{A}|^2 |\vec{B}|^2$, also give the geometric interpretation of this equation. (CLO1, PLO1, 5 MARKS)

(B) A force $\vec{F} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ Newtons is applied at a point whose position vector from the pivot is $\vec{r} = 2\hat{i} - 3\hat{j} + \hat{k}$ meters. Calculate the torque and its direction about the pivot point, also find the angle (θ) between $\vec{r}$ and $\vec{F}$. (CLO3, PLO3, 2.5 MARKS)

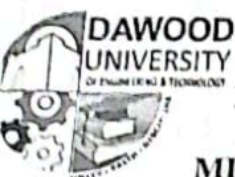
Q.NO. 3 (A) Explain briefly Newton's second law of motion and impulse, also derive the equation for impulse by using Newton's second law of motion. (CLO2, PLO2, 5 MARKS)

(B) A boy kicks on an object of half kg so that its position (in meters) as a function of time (in seconds) is given by $x = 2t^3 - t$. Find the impulse delivered to the object between $t = 0$ s and $t = 1$ s. (CLO3, PLO3, 2.5 MARKS)

Q.NO. 4 (A) Explain briefly the derivation of the relation for electric field intensity due to an electric dipole at a distance 'x' from its centre? (CLO2, PLO2, 5 MARKS)

(B) The electric field at a point 10 cm from the centre of a dipole (dipole consists of two charges, $+2 \mu\text{C}$ and $-2 \mu\text{C}$) is $5 \times 10^4 \text{ N/C}$. Calculate the dipole moment of the dipole, separation of dipole and force on a test charge 1 nC placed at point 10 cm from the centre of same dipole. (CLO3, PLO3, 2.5 MARKS)

THE END



DATED 27-October-2025 TIME ALLOWED 90 MINUTES

MAX. MARKS 30

Student Name: Syed Sheharyar Tariq

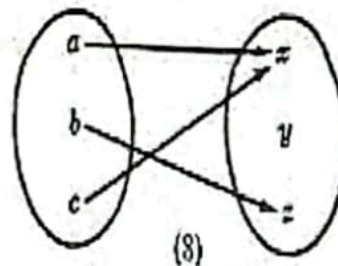
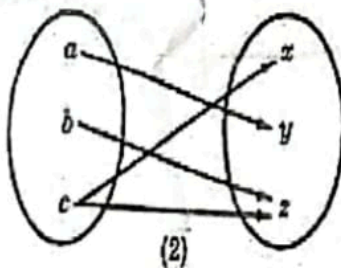
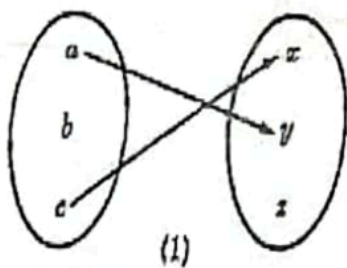
Student Id Number: 25F-BSAI-077

Instructions:

1. This paper contains 04 questions.
2. Use of Calculator is allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationery is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned back to invigilator before leaving exam hall.

NOTE: ATTEMPT ALL THE FOLLOWING QUESTIONS.

Question # 01: Distinguish whether the given relations are functions? If not, justify your answer otherwise mention the type of function.
(CLO-1, PLO-1, C2, 7 Marks)



Question # 02: Compute the following limits

a. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$

b. $\lim_{x \rightarrow 2} f(x)$ is defined as $f(x) = \begin{cases} x^2 - x & x \leq 2 \\ \sqrt{x + 9} & x > 2 \end{cases}$

(CLO-1, PLO-1, C2, 8 Marks)

Question # 03: Function is defined as:

$$f(x) = \begin{cases} \frac{x^3 - 27}{x - 3} & x \neq 3 \\ \frac{9}{2} & x = 3 \end{cases}$$

Determine whether $f(x)$ differentiable and continuous at $x=3$?

(CLO-2, PLO-2, C3, 8 Marks)

Question # 04: Solve each of the following inequalities

a. $|x| + |x - 1| > 0$

b. $\frac{x^2 - 2}{1 - 2x} > 1$

(CLO-2, PLO-2, C3, 7 Marks)

FUNCTIONAL ENGLISH (THEORY)

DATED 01-11-2025

TIME ALLOWED 45 MINUTES

MAX. MARKS 15

Student Name : _____

Student Id Number : _____

Instructions:

1. This paper contains 02 questions.
2. Use of Calculator is not allowed
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationary is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned back to invigilator before leaving exam hall.
- 10.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.1. Read the passage carefully and then attempt all questions.

[C-2, CLO-01, PLO-07, 09 Marks]

Youthquake in Pakistan

(Adapted from Dawn Editorial)

In recent years, Pakistan has seen a big change in the way young people think and act. Most of the country's population is under 30 years old. This means that young people now have great power to shape the future.

Today, political discussions are not only happening in government offices or big meetings. They are also happening online, on social media, in universities, and in small groups of friends. This quiet but strong change shows how young people are using new ways to express their opinions.

This generation has **grown up** in hard times, during wars, economic problems, and limits on free speech. **They prefer fast, honest, and direct communication.** A short video, a funny meme, or a simple hashtag can spread a message faster than a long speech or a big rally.

Young people quickly move to new digital platforms whenever old ones are blocked. **They are smart and adaptable.** Every time someone tries to silence them, **they find another way to speak up.** This can bring both good and bad results.

If the voices of young people are ignored, they may feel hopeless or turn to anger. But if they are listened to and given space to participate, they can **bring up** new ideas and energy to strengthen democracy in Pakistan. Therefore, schools, universities, and leaders should help young people take part in discussions and decision-making. Working with them, not against them, will make the system stronger.

- i. What is the main idea of the passage "Youthquake in Pakistan"? /01 mark
- ii. Where are young people in Pakistan mostly sharing their opinions today? /01 mark
- iii. Why does the writer mention in the third paragraph that young people prefer "*fast, honest, and direct communication*"? /02 marks
- iv. In the fourth paragraph, the writer mentioned, young people "*find another way to speak up.*" Why do you think freedom of expression is important for them? /02 marks

- v. How are young people's ways of expressing opinions today different from those of older generations? Write at least three differences. /03 marks

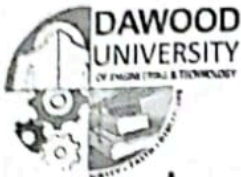
[C-2, CLO-01, PLO-07, 06 Marks]

Q.2a Infer the meanings of the following phrasal verbs as used in the passage. Use contextual clues from each paragraph to support your answer: /3 marks

- a. grown up. (third paragraph)
- b. speak up (fourth paragraph)
- c. bring up (fifth paragraph)

Q.2b. What can teachers and leaders do to make young people feel more involved in improving their country? Provide at least three suggestions. /3 marks

X-----The End-----X



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

MIDSEMESTER EXAMINATION OF 1ST SEMESTER, 1ST YEAR 2025 BATCH (FALL SEMESTER)
OF BS ARTIFICIAL INTELLIGENCE SECTIONS (A1, A2, & A3) OF 2025

SUBJECT: INTRODUCTION TO INFORMATION & COMMUNICATION
TECHNOLOGIES

DATED: 30-10-2025

TIME ALLOWED 45 MINUTES

MAX. MARKS 15

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains **02** questions.
2. Use of Calculator is allowed/Not allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationery is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned to invigilator before leaving exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q. No. 1 a) **Describe** the basic components of computer systems in terms of hardware..
(CLO-1, C-1, 3.5 Marks)

Q. No. 1 b) **Describe** the specification of Super computers and determine applications of Super computers in our daily lives.
(CLO-1, C-1, 04 Marks)

Q. No. 2 a) **Express** the term Computer Based Information systems and also list its elements.
(CLO-2, C-3, 04 Marks)

Q. No. 2 b) **Construct** the flow chart if he got 80 marks then shows the Your name got A- marks.
(CLO-2, C-3, 3.5 Marks)

-----THE END-----



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

MIDSEMESTER EXAMINATION OF 1ST SEMESTER, 1ST YEAR (FALL SEMESTER)
OF BS ARTIFICIAL INTELLIGENCE SECTIONS (A1, A2 & A3), OF 2025

SUBJECT: ISLAMIC STUDIES / ETHICS

DATED: 31-10-2025

TIME ALLOWED 45 MINUTES

MAX. MARKS 15

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 02 questions.
2. Use of Calculator is allowed/Not allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationery is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned to invigilator before leaving exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.NO. 1. Explain the qualities of Muslims as described in "Surah-e-Mominoon."

(CLO: 01, Level: 02, PLO: 06) (Marks: 07)

Q.NO. 2. Define "Islam and Science." Elaborate with related examples.

(CLO: 02, Level: 02, PLO: 08) (Marks: 08)

ETHICS

Q.NO. 1. Explain in detail the meaning of "5 Ks." Also Elaborate the concept of God in Sikh Religion.

(CLO: 01, Level: 02, PLO: 06) (Marks: 07)

Q.NO. 2. Interpret the festivals of Holi and Vasaki.

(CLO: 02, Level: 02, PLO: 08) (Marks: 08)

THE END



MIDSEMESTER EXAMINATION OF 1ST SEMESTER, 1ST YEAR 2025 BATCH (FALL SEMESTER) OF BS ARTIFICIAL INTELLIGENCE SECTIONS (A1, A2, & A3) OF 2025.

SUBJECT: PROGRAMMING FUNDAMENTALS (THEORY)

DATED: 28-10-2025

TIME ALLOWED 90 MINUTES

MAX. MARKS 30

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 04 questions.
2. Use of Calculator is allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationery is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned to invigilator before leaving exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

(1 * 7 = 7)

Q # 1: Select the correct option from the following choices.

- i. Which Word processing programs, spreadsheet programs, e-mail programs, Web browsers, and game programs belong to what category of software?
A) System Software B) Application Software C) Both A & B D) None of these
- ii. Words that have special meaning in a programming language are called _____.
A) Variables B) Operators C) Keywords D) Functions
- iii. A variable must be _____ before it can be used in a program.
A) Declared B) Assigned C) Executed D) Printed
- iv. Which of the following is not a valid assignment statement?
A) total = 9; B) 72 = amount; C) letter = 'W'; D) None of these
- v. Assume x is an integer variable. If the user enters 3.14 in response to the following programming statement, what will be stored in x?
A) 3.14 B) 0 C) 3 D) An error message is displayed.
- vi. Which loop should you use in situations where you wish the loop to repeat until the test expression is false, but the loop should execute at least one time?
A) for loop B) do-while loop C) while loop D) infinite loop
- vii. The trailing else in an if-else-if statement has a similar purpose as the _____ section of a switch statement.
A) case B) switch C) break D) default

Q # 2: Answer the following questions:

(2.5 * 3 = 7.5)

- a. What is the purpose of the following escape sequences?
 - i. \n
 - ii. \t
 - iii. \'
 - iv. \"
 - v. \\

- b. What is software/program, and what are the different types of software?
- c. What is the difference between high-level and low-level programming languages? (2.5 * 3 = 7.5)

Q # 3: Answer the following questions:

- a. Write a C++ program that determines if a student has passed a course. The passing criteria are:

- ✓ The student must have scored at least 50 marks and have attended at least 75% of classes.

(Note: Take marks and attendance as input from the user.)

- b. What is the output of the following code?

```
#include<iostream>
using namespace std;

int main()
{
    float f1 = 4.0, f2 = 2.0;

    cout << f1 % f2;

    return 0;
}
```

- c. State the order of evaluation of the operators in each of the following C++ statements and show the value of x after each statement is performed.

- i. $x = 7 + 3 * 6 / 2 - 1;$
- ii. $x = 2 \% 2 + 2 * 2 - 2 / 2;$
- iii. $x = (3 * 9 * (3 + (9 * 3 / (3))));$

Q # 4: Answer the following questions:

(4 * 2 = 8)

- a. Write a menu-driven C++ program that interacts with the user to perform temperature conversions. The program should:

- i. Prompt the user to input a temperature value in Celsius.
- ii. Display a menu asking whether the user wants to convert this temperature to Fahrenheit or Kelvin.
- iii. Accept user input for the choice, where pressing 'F' or 'f' indicates conversion to Fahrenheit, and 'K' or 'k' indicates conversion to Kelvin.
- iv. Perform the selected conversion using the appropriate formula:
 - Fahrenheit = ((Celsius * 9.0) / 5.0) + 32
 - Kelvin = Celsius + 273.15
- v. Display the converted temperature along with a descriptive message.

(Note: You may implement the decision logic using either if-else-if statements or a switch statement, based on your preference.)

- b. When should you choose each type of loop (for, while, or do-while) in programming? Explain the scenarios in which each loop is most appropriate, providing examples to illustrate your points.

THE END